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solar photovoltaics

a brief history of technologies

In this issue's "History" column, we examine the utilization of solar energy from a variety of historic perspectives and the evolution of different types of photovoltaic (PV) cells. We welcome Sayan Kumar Nag and Tarun Kumar Gangopadhyay of the Department of Electrical Engineering, Techno Main Salt Lake Kolkata, West Bengal, India, who join us for their first time to our "History" pages.

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Associate Editor, "History"

IN THE PRESENT CENTURY, SOLAR energy has emerged as an important source of nonconventional energy to meet the energy demand for overall development of a nation. The use of solar energy for human development is not a new discovery but instead is a century-old tradition. As the demand for clean energy sources increases, the importance of the development of efficient photovoltaic (PV) cells is in demand. Here we examine the utilization of solar energy in the initial stage, the rise of PV development in the present era, and different kinds of PV cells with their merits and demerits.

Solar Energy—The Initial Years

It is a common belief among many people that the use of solar energy is a phenomenon of modern times either as a repercussion of the 1970s oil embargo, a byproduct of the 20th-century space race, or both. Nevertheless, it isn't all the truth. When considering energy sources in the modern world, the first thing that generally comes to mind is traditional energy sources such as coal, oil, and gas. But biomass from wood, dried animal manure, and peat has traditionally been used as fuel for ancient cultures. Obviously, like other forms of renewable energy, both traditional and modern biofuels originally generated from the Sun. The biofuels act like early storage units of solar energy and are only useful during daylight hours. Ancient civilizations, such as Native Americans, Babylonians, Persians, ancient Hindus, and Egyptians, held the Sun in high regard to the point of worshipping it. However, it was actually the Greeks and Romans who first realized the Sun's power as a source of free energy. The first known way to capture the Sun's rays occurred around 600 BCE, when wood was ignited by focusing the rays onto it through magnifying lenses. An illustration of this is shown in Figure 1.

figure 1. An illustration depicting the Sun's rays being focused to start a fire in 1658. (Source: J.A. Comenius, *Orbis Sensualium Pictus*, Michael Endter, 1658.)

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Few civilizations focused on developing new ideas or technologies unless the need for development grew. Greece experienced a shortage of fuel in the fourth century BCE, so it became innovative and started with the idea of providing heat and light since it was very important for the survival of the community. Greece was the first civilization to consciously use solar energy in its dwellings for reasons other than religious worship, as early as 400 BCE, implementing passive solar design into homes.

As shown in Figure 2, passive solar design ensures protecting the north

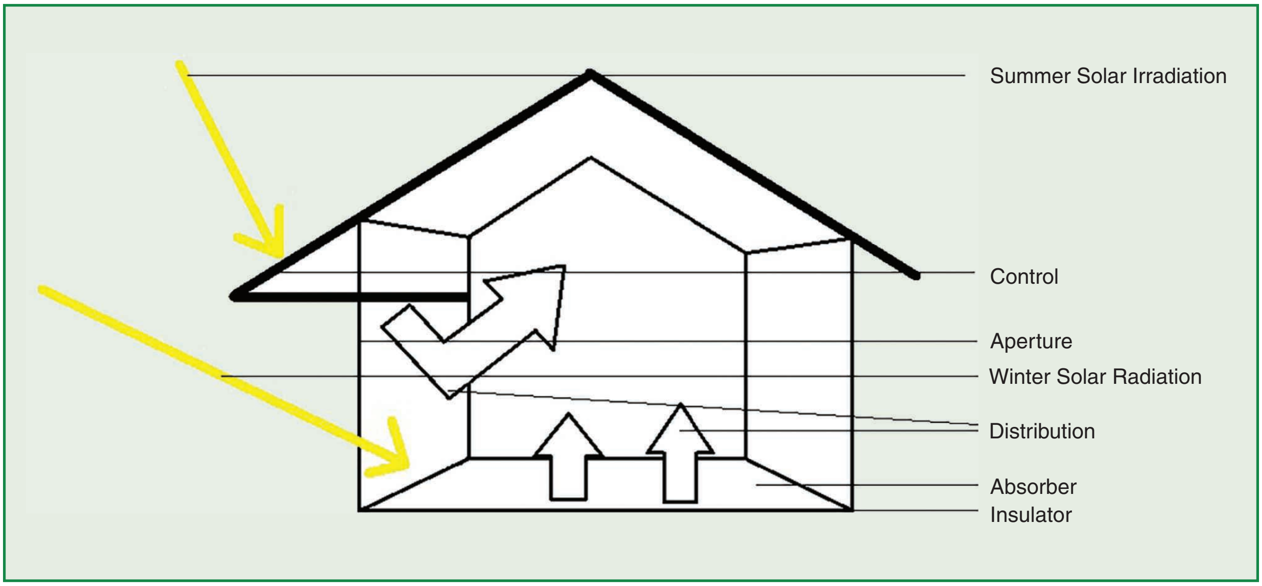

figure 2. A passive solar design to protect north side from wind and rain.

figure 3. A photo of ancient Roman baths.

face of a building from external elements (wind, rain, and so on). On the south side, the solar radiation during the winter was open, but was shaded in the summer. This simple principle was widely accepted and implemented not only in private houses but also in public buildings and the world famous Roman baths. A picture of a Roman bath is shown in Figure 3.

Photovoltaic Development—A New Era

For centuries, we have used the power of the Sun in several ways. But it wasn't until the 1800s that scientific breakthroughs allowed us to fully exploit the potential of this free and abundant fuel source. A French physicist named Edmund Becquerel (Nobel laureate) published his experiments conducted with wet cell batteries in 1839, which led to the transition to solar energy use.

In the 1800s, there was further support for this discovery, but it took until the mid-20th century for proof to be made. A publication by William G. Adams and Richard E. Day considered the effects of sunlight on selenium, and Charles Edgar Fritts developed a very inefficient prototype cell (1%–2% efficiency) during the late 1880s that looked like a typical cell in use today. A typical diagram of a modern PV cell is shown in Figure 4.

To calculate the efficiency of PV cells, one must measure the amount of electricity generated using the total potential energy received by the PV module. In many ways, the prototype solar cell resembles today's modern cells. Beneath it was a glass sheath with a fine chunk of gold wire sandwiched between the glass and a thin layer of selenium. This was the first model for further improvement by scientists, but it would take decades before efficiency improved and a true understanding of the causes of early low power is reached.

A lack of interest, a lack of knowledge, or the low price of fossil fuels may be behind the little development that occurred in the half-century after the initial discovery. The only notable event in the first half of the 20th century that followed the same trend was Max Planck's new idea, his light quantum hypothesis.

An explanation of the photoelectric effect was published by Albert Einstein in March 1905. Five years earlier, Planck had shown that atoms in cavities can only absorb or emit radiation from discrete "quantum," which is why the energy of each proton is an integer multiple: frequency times new default constant. Planck thus solved the problem of blackbody radiation.

Planck believed that his conception of quantum was nothing more than a

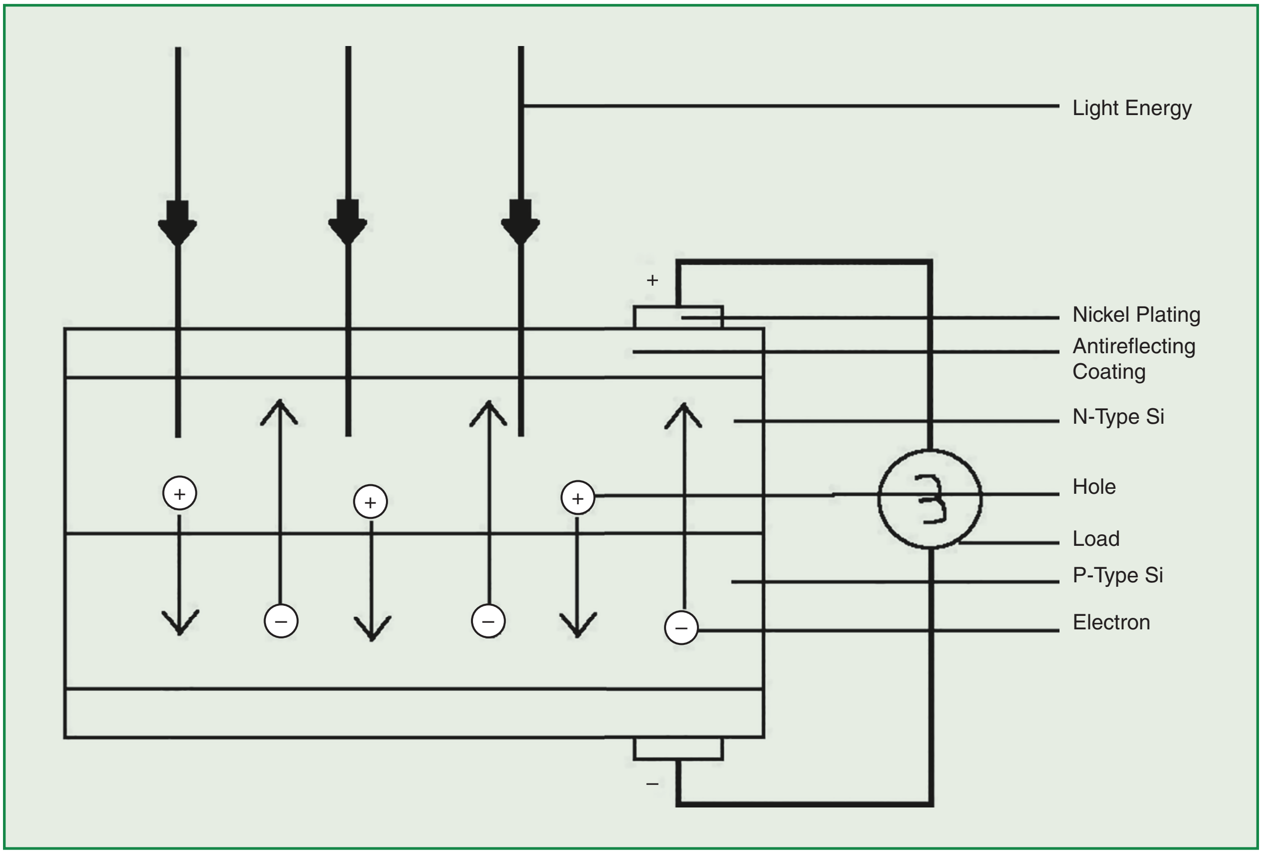

figure 4. A schematic of a PV cell. Si: silicon.

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mathematical "trick" to fit the theory into experimentation. But Einstein extended Planck's quantum into light itself (Planck assumed that only the vibrations of atoms were quantized). Einstein said that light is a beam of particles whose energy is related to frequency according to Planck's formula. A beam of light collides with atoms upon impact with the metal. If the photon's frequency is sufficient to drop the electrons, the collision causes the photoelectric effect. As a particle, light carries energy proportional to the frequency of the wave. As a wave, it has a frequency determined by the energy of the particle. This work earned Einstein the 1921 Nobel Prize in Physics.

figure 5. The space satellite *Vanguard 1* with its six solar cells attached. (Source: National Space Science Data Center, NASA, Vanguard 1 with its six solar cells attached, 2007.)

Researchers at Bell Laboratory found that by mixing arsenic with silicon and a thin coating on the cell with boron, cell efficiency of 6% (US$250 per watt versus US$3 per watt for coal) can be achieved. Perhaps even if their findings were separate from transistor technology and expertise, it would not have been known to scientists who started the energy supply revolution that would be needed globally within 20 years. Despite advances, the current product was too expensive for use on the ground. However, for space power applications, it was a perfect solution as there were no other alternatives and, despite the demerits, it was still the best solution.

The first of the many solar cells that powered the space machinery were deployed on the *Vanguard I* space satellite (Figure 5) in 1958 and operated until decommissioned in 1964. As an endless pollution-free power supply, the tremendous success of PV cells has secured a place for these cells. In spite of their efficiency and cost, PV cells have become a ubiquitous power source in the aerospace industry due to their success as pollution-free infinite sources of electricity.

After the French discovery in the early 19th century, the United States led the development of solar cells. Until 1990, it was a leader in the market, research and development, and implementation. Nevertheless, at the end of the 20th century, this domination changed as it moved and was divided between Europe and Japan. Over the past decade, the world has reached a tremendous milestone integrating solar power into 1 million households. During this period, Japan and Europe, especially Germany,

table 1. Summary highlights of solar PV technologies of multijunction cells, crystalline modules, thin films, and some new solar modules.

Solar Technology	Characteristics	Efficiency in 2019 (%)	Advantages	Drawbacks
Monocrystalline	Monocrystalline silicon is a pure material with a continuous, unbroken crystalline lattice structure that has minimal or no impurities.	26.7	1) Longevity 2) Low installation cost 3) Nonhazardous	1) High cost 2) Fragility 3) Silicon wasted in the Czochralski process
Polycrystalline	Numerous monocrystalline silicon grains are used to manufacture polycrystalline silicon.	22.3	1) Simple production process 2) More heat-tolerant than silicon-based panels	1) Generally less efficient than monocrystalline 2) Space efficiency is lower
CIGS	Solar energy is converted into electricity by semiconductor layers in CIGS.	23.4	1) The material is less or nontoxic, unlike Cadmium telluride (CdTe) 2) Better at resisting heat compared to Si-based panels	1) A less efficient alternative to monocrystalline panels
CdTe	It is based on a superstrate comprising of successive layers of soda lime glass, indium-tin oxide, Cadmium sulfide, and CdTe back contact.	21.0	1) Solar energy is absorbed at shorter wavelengths 2) Cadmium is abundant, resulting in lower manufacturing costs	1) Cadmium is highly toxic 2) Old CdTe panels may pose a disposal problem 3) Less efficient compared with crystalline silicon
Amorphous Silicon	It is made of silica atoms are arranged in thin, homogeneous layers.	13.4	1) Multijunction device capacity 2) Easy fabrication 3) High optical absorption coefficient 4) The material is less toxic compared to CIGS or CdTe 5) Flexible and less susceptible to crack	1) Low efficiency
Dye-sensitized solar cells	It includes the electrode film layer, the conductive transparent conductive oxide, the counter electrode layer, and redox electrolyte layer.	11.1	1) Has a lower cost 2) Can operate in dim light and from a wider angle 3) Durable and mechanically robust	1) Volatile organic solvents make up electrolyte liquid 2) Cannot be used with large-scale cases with high cost and efficiency requirements
Perovskite solar cell	It uses active light harvesting layer based on the ABX_3 crystal structure, also known as *perovskite*.	20.9	1) Cost-effective fabrication 2) Inexpensive to scale up 3) Less space required for installation 4) High efficiency	1) Material degradation occurs quickly when exposed to heat, moisture, or snow 2) The substance is toxic by nature
Polymer	It contains a layered structure consisting of a transparent front electrode, an active layer, and a back electrode.	3–8	1) Easy to size and shape due to its light weight and flexibility 2) Easily storable and transportable 3) Ecologically friendly due to being organic in nature	1) Extremely low efficiency 2) Degradation is faster when exposed to outdoor conditions

Source: T.M. Razykov, C.S. Ferekides, D. Morel, E. Stefanakos, H.S. Ullal, H.M. Upadhyaya, “Solar photovoltaic electricity: Current status and future prospects,” Solar Energy, vol. 85, no. 8, pp. 1580–1608, Aug. 2011. (https://doi.org/10.1016/j.solener.2010.12.002)

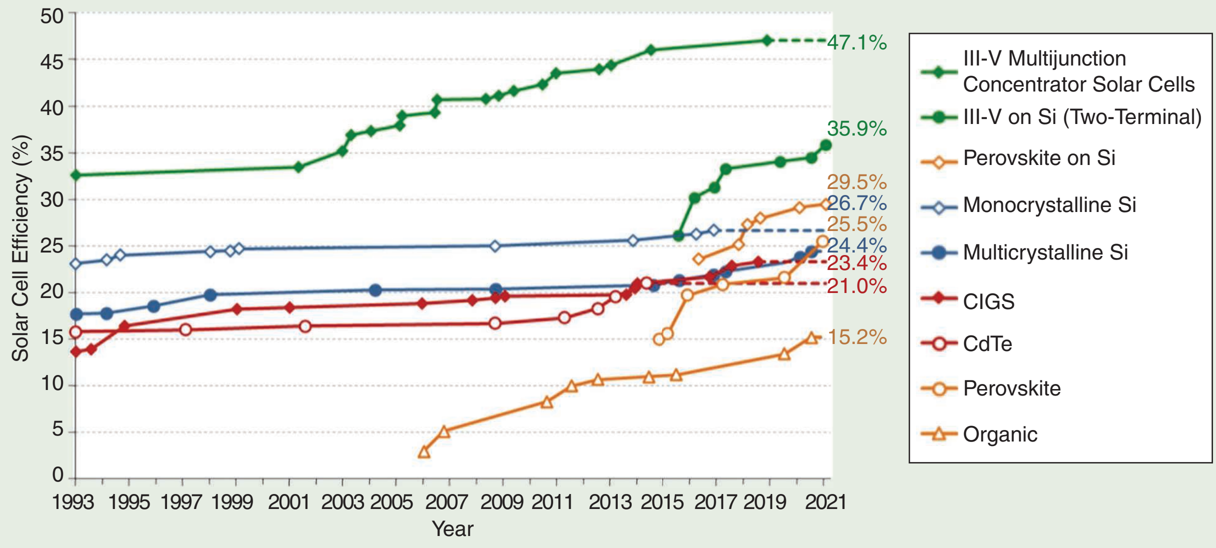

figure 6. The development of laboratory solar efficiencies of multijunction cells, crystalline modules, thin films, and some new solar modules over time. CIGS: copper indium gallium selenide; CdTe: cadmium telluride. (Source: Fraunhofer Inst. for Solar Energy, "Photovoltaics report," Jul. 27, 2021; used with permission.)

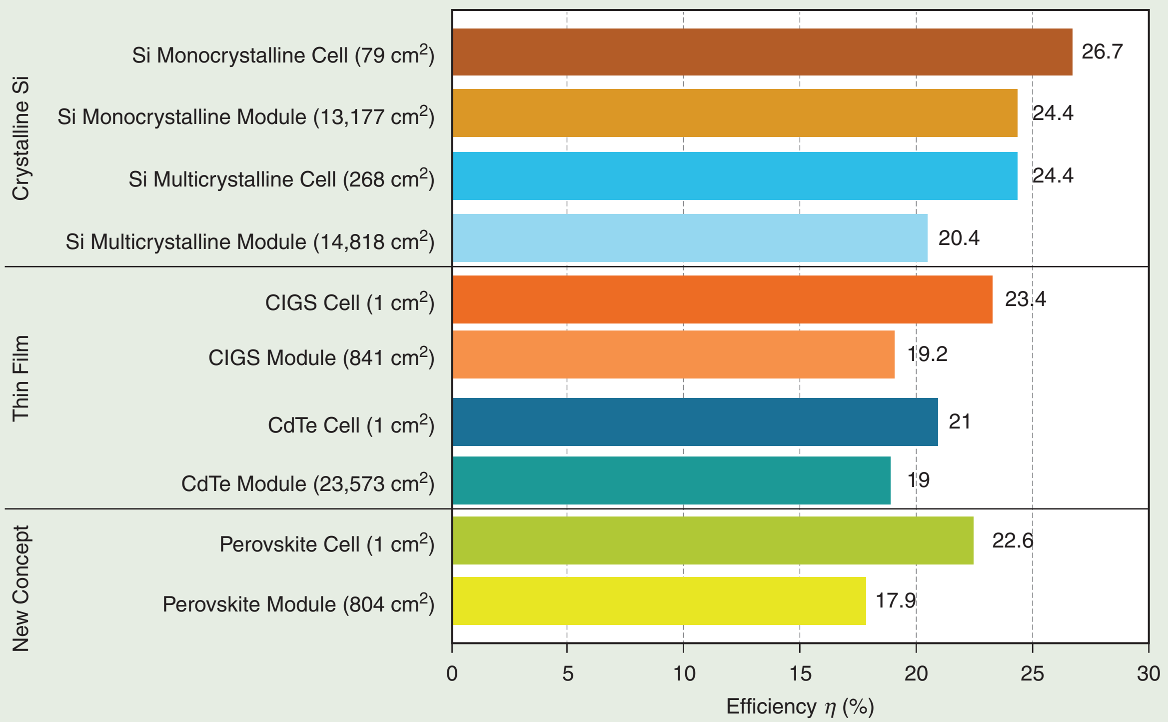

figure 7. The solar PV efficiencies of multijunction cells, crystalline modules, thin films, and some new solar modules. (Source: Fraunhofer Inst. for Solar Energy, "Photovoltaics report," Jul. 27, 2021; used with permission.)

introduced government subsidies, raised public awareness, and invested in research and development. In the 1990s, Japan saw a ten-fold increase in the market as Germany initially weakened, but as a result of modifying subsidies, production increased forty-fold, surpassing Japan's success. Their example has been followed by other European countries, such as Spain, and have achieved similar growth rate.

There have been a considerable number of levels of development in the PV market over the past several decades, including research and product development for tasks like lighting, desalination, and pumping. Thus, it has become interesting not only for scientists and engineers but also for the general public. Solar energy is now an inexpensive, highly efficient, and clean alternative to fossil fuels as well as more easily accessible power supplies to meet an ever-increasing demand.

Solar PV Modules—Performance and Efficiency

Solar cell performance depends on the fill factor and conversion efficiency. Mathematically, the fill factor is defined as the ratio of the product of the maximum current value (I_m) and the maximum voltage value (V_m) divided by the product of the short circuit current (I_{SC}) and the open circuit voltage (V_{OC}). The value of the conversion efficiency is the ratio of P_m (the maximum power generated by the panel under standard test conditions) and P_{in} (the solar radiation, i.e., radiant energy emitted from the Sun).

According to the data obtained from the solar efficiency (Table 1), the efficiencies of multijunction cells and crystalline modules outperform those

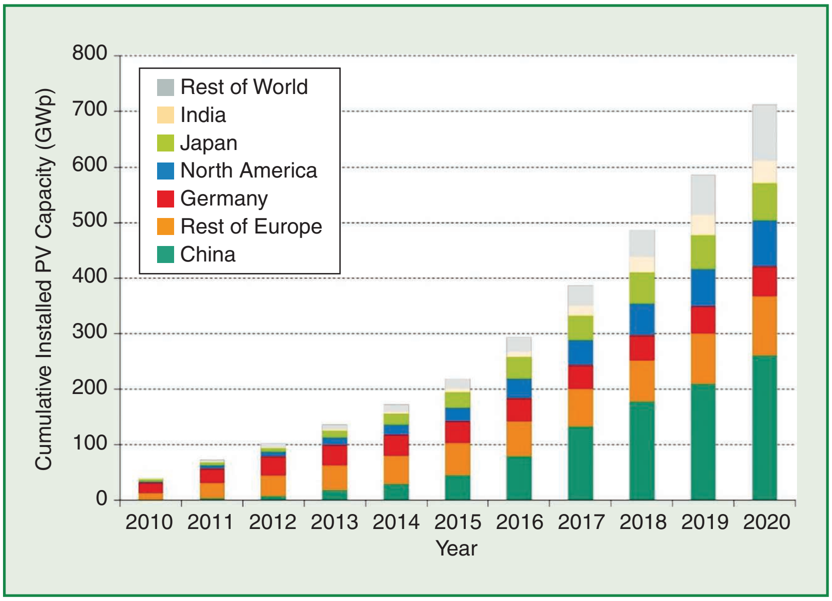

figure 8. The global cumulative PV installation by region. (Source: Fraunhofer Inst. for Solar Energy, "Photovoltaics report," Jul. 27, 2021; used with permission.)

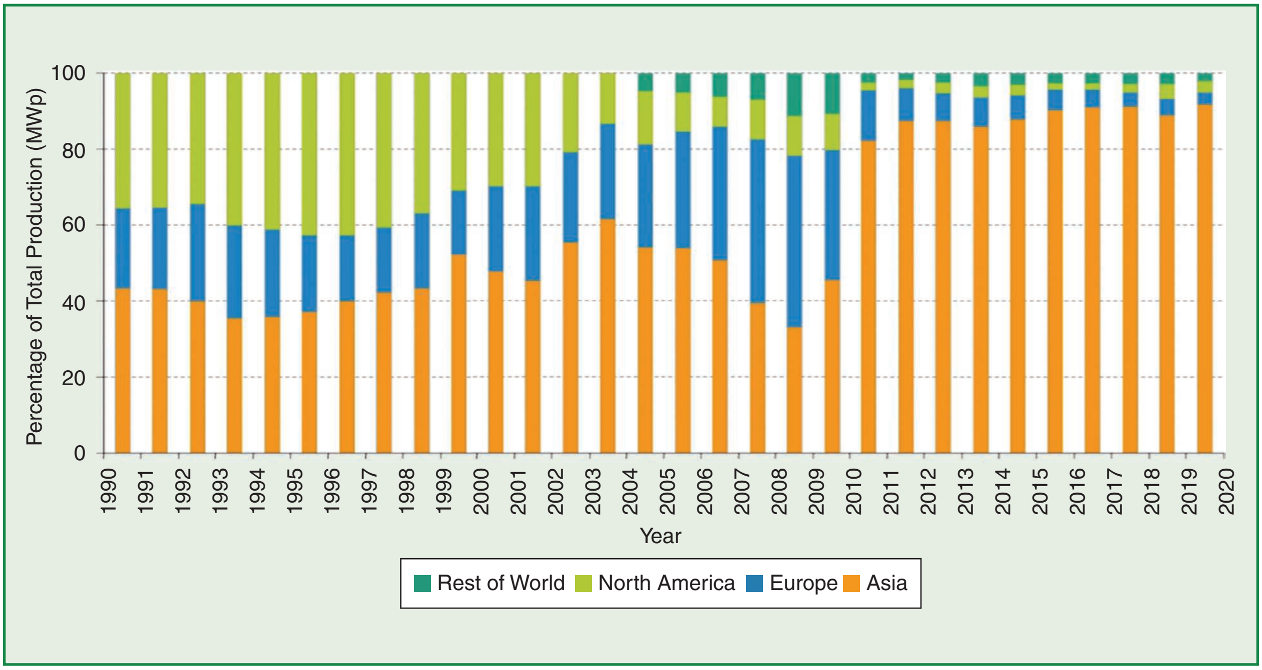

figure 9. PV module production by region, 1997–2020. (Source: Fraunhofer Inst. for Solar Energy, "Photovoltaics report," Jul. 27, 2021; used with permission.)

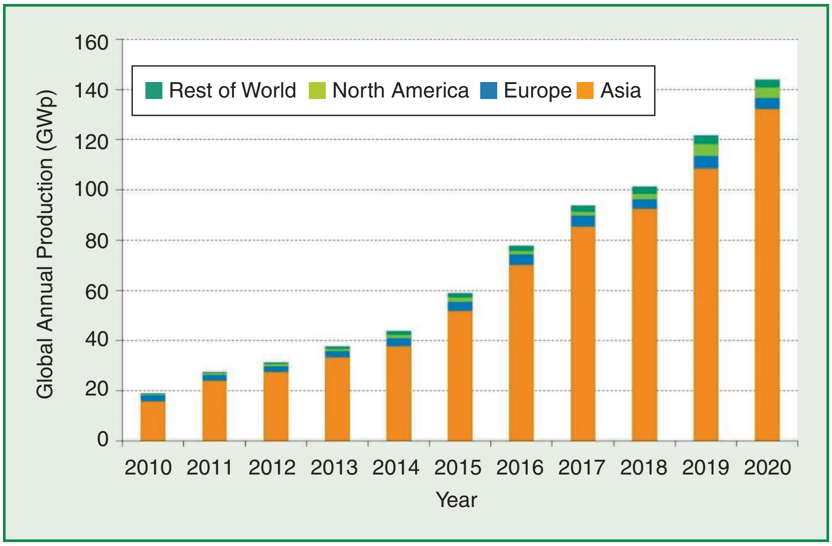

figure 10. The global annual PV module production by region. (Source: Fraunhofer Inst. for Solar Energy, “Photovoltaics report,” Jul. 27, 2021; used with permission.)

of thin films and some new solar modules. Figures 6 and 7 illustrate the data. Monocrystalline solar cell efficiency is about 4.3% higher than that of polycrystalline, while copper indium gallium selenide (CIGS) is about 2.4% higher than that of cadmium telluride (CdTe). Table 1 highlights the key features and efficiency levels of different solar PV technologies as well as the pros and cons of each implementation.

Solar PV Modules—Production and Installation

At the end of 2020, China had the most installed solar PV power, followed by Europe and North America, as shown in Figure 8. The chart in Figure 9 shows the global production rate of

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table 2. Annual gross generation of power by source in India.									
Fuel Category	**2013**	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021******
Coal*	145,273	164,636	185,173	192,163	197,171	200,704	204,224	205,135	209,810
Gas	21,782	23,062	24,509	25,329	24,897	24,937	24,937	24,955	24,924
Diesel	1,199	1,199	993	838	838	638	510	510	510
Nuclear	4,780	5,780	5,780	6,780	6,780	6,780	6,780	6,780	6,780
Hydro**	44,335	45,323	47,057	48,858	49,779	49,992	50,047	50,382	51,351
Wind	21,043	23,354	26,777	32,280	34,046	35,626	37,279	37,694	40,083
Solar	2,632	3,744	6,763	12,289	21,651	28,181	32,578	34,628	49,347
Biomass***	7,510	7,805	8,110	8,296	8,839	9,242	9,946	10,023	10,610
Total	248,554	274,903	305,162	326,833	344,001	356,100	365,891	370,127	393,415

*Includes lignite. **Includes small hydropower. ***Includes biomass power/cogeneration and energy-from-waste.
****Preliminary data based on 31 December 2021.
[Source: Central Electricity Authority, CEA (2019a), "All India installed capacity," January 12, 2022 (https://www.cea.nic.in).]

solar modules over 20 years. Europe, Asia, and North America have made significant investments in the production of solar modules, while China has made a significant increase in production. Data in the rest of the world indicates a gradual increase in production compared to 2010. Figure 10 shows that over the past 10 years, Asia has continued to increase solar PV production toward the goal of reducing its annual carbon footprint and environmental pollution. It has been followed by other Asian countries like India and Japan, as shown in Figure 8.

India's installed generation capacity is comprised of 365 GW, coal (55.8%), hydro (13.7%), wind (10.1%), solar PV (8.8%), natural gas (6.8%), bioenergy and waste (2.7%), nuclear (2%), and oil (0.1%). India has rapidly increased its installed solar capacity from 2.63 GW in 2013 to 32.5 GW in 2019. Even though renewable sources (wind, solar, and bioenergy) accounted for 8% of generation in 2017, they accounted for 21.8% of installed capacity. Annual gross generation of power by source in India is shown in Table 2. There is a huge increase in solar power plants as concluded from Table 2.

Conclusion

As the research and development of PV cells are in progress, their efficiency is increasing and prices are coming down. As discussed earlier, solar cells are being developed that have almost 40% efficiency. We also know that when demand and production go up, prices come down. Therefore, we can conclude that in the coming future, solar cells can emerge as the single largest supply system of energy just as coal emerged in the 19th century during the Industrial Revolution. The future looks bright for solar PV technologies.

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For Further Reading

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